Chapter 1: Numerical Methods

We introduce a locally mass conserving numerical framework in this chapter. A conforming mixed finite element discretization (MFEM) is proposed to solve the static Darcy-Stokes coupled methcnics system. The normal flux are captured at the edges of element by the MFEM scheme. The coupled weak form is also adjusted to be locally mass conservative. We solve the transport system with finite volume method (FVM), which is naturally a locally mass conservative scheme.

In this chapter, we first introduction the mesh used in our computation. We present two stable pair for Darcy and Stokes individually equipped with the direct serendipity space. We then apply the defined discrete space to discretize the coupled system. After that, we discuss a noval nonlinear WENO weighting scheme for FVM. Error estimate will be given to MFEM and FVM scheme. We provide a detailed coupling strategy combining MFEM, FVM and eutectic phase package in the end of this chapter.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$. Here, $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. Let $\mathcal{T}_h$ be a conforming mesh of quadrilaterals covering Ω. The quadrilateral $E \in \mathcal{T}_h$ is referred as element in finite element context, while referred as cell in finite volume context. Each quadrilateral E is assumed to be strictly convex, not degenerate into a triangle, line or point. For ease of implementation, we use a logically rectangular mesh, where each element(cell) is indexed using Cartesian coordinates $\{i, j\}$. We show a reference element(cell) $\hat{E} = [-1, 1]^2$, an unscaled element $\tilde{E}$ and a scaled general quadrilateral element(cell) $E/H = \tilde{E}$ in Figure 1.1. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to \tilde{E}$ mapping the vertices of $\hat{E}$ to the vertices of $\tilde{E}$.

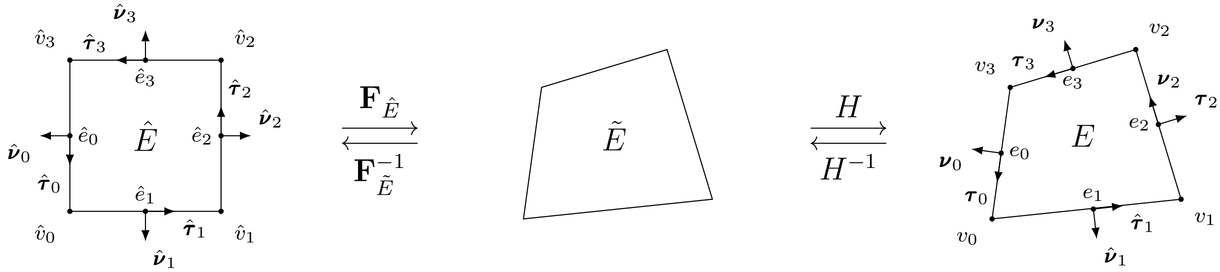


Figure 1.1: An exhibition of a reference square and general non-degenerate quadrilateral element(cell) before and after scaling.

The edges of element E are denoted by e_i, $i \in \{0, 1, 2, 3\}$ in the counterclockwise direction. We begin counting from zero, first to align with the indexing convention of C++, and second to facilitate representation in terms of modulo arithmetic, which also start from zero. The vertices are represented as $v_i = e_i \cap e_{i+1}$, and $i \equiv i \pmod{}4$. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outerwards the quadrilateral. The tangent $\boldsymbol{\tau}_i$ is generated by rotating $\boldsymbol{\nu}_i$ counterclockwise by right-hand rule.

In Figure 1.2, we illustrate a full logically rectangular qusi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distort the interior vertex of a square mesh. Note that vertex on the boundary $\partial\Omega$ remain untouched.

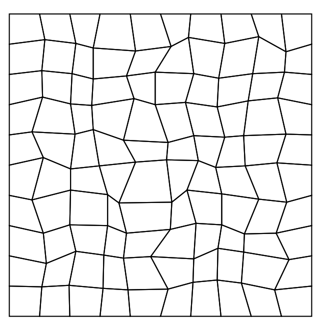


Figure 1.2: An exhibition of a full logically rectangular qusi-uniform quadrilateral mesh $\mathcal{Q}_h$.

1.1.1 Shape Regularity

We give an official definition of uniform shape regularity of a quadrilateral mesh.

Definition 1.1. For any quadrilateral $E \in \mathcal{T}_h$, we can get a set of four triangles $T_i, \quad i \in \{0,1,2,3\}$. We form the triangle T_i by excluding vertex v_i from the quadrilaterl E, and use the remaining three vertex. We define the following parameters as

$$\begin{aligned} h_E &= \text{diameter of E}, \\ \rho_E &= 2 \min_{1 \le i \le 4} \{\text{diameter of largest circle inscribed in } T_i\}. \end{aligned} \tag{1.1}$$

A mesh partion $\mathcal{T}_h$ is considered to be uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that the ratio

$$\rho_E / h_E \ge \sigma_* > 0, \tag{1.2}$$

for all quadrilaterals $E \in \mathcal{T}_h$.

1.2 Direct Serendipity Space

Serendipity element have the same accuracy of approximation with classical tensor product element while use smaller number of degrees of freedom. However, the classical serendipity space suffer from poor approximation on quadrilaterals. We define local shape functions directly on quadrilaterals according to Arbogast et al. (2022). H^1 and $H(\text{div})$ conforming direct mixed finite elemenent space are constructed with the direct serendipity function space.

To begin with, we introduce some notations and distance auxilliary functions. Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ can be calculated as

$$\text{dim}\mathbb{P}_r(\mathbb{R}^d) = \binom{r+2}{2} = \frac{(r+2)!}{r!2!}. \tag{1.3}$$

We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{1.4}$$

where $\mathbf{x}_i^* \in e_i$ is any point on the edge e_i. We pick $\mathbf{x}_i^* = \mathbf{x}_{v,i}$ for simplicity. The value of distance function λ_i keeps positive as long as $\boldsymbol{x}$ is in the interior of the quadrilateral. We define two additional distance functions giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\nu_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by counterclockwise rotating of the vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{d,i}^*) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 0, 1, \tag{1.5}$$

where $\mathbf{x}_{d,i}^*$ is any point on the diagonal line d_i. Here, we use the corner point v_i for simplicity, $\mathbf{x}_{d,i}^* = \mathbf{x}_{v,i}$, where $\mathbf{x}_{v,i}$ denotes the coordinates of corner v_i.

Now we present set of nodal basis functions in first and second order direct serendipity space $\mathcal{DS}_1$ and $\mathcal{DS}_2$ defined by Arbogast et al. (2022).

1.2.1 Direct Serendipity Space $\mathcal{DS}_2$

We supplement polynomial space $\mathbb{P}_r(E), r \geq 2$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$, spanning the supplemental function space as $\mathbb{S}_r^{\mathcal{DS}}(E) = \text{span}\{\phi_{s,1}, \phi_{s,2}\}, \quad r \geq 2$. We present the second order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}_2^{\mathcal{DS}}(E). \tag{1.6}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{0, 1\}, \tag{1.7}$$

where $i \equiv i \pmod 4$. We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.8}$$

which satisfies the conditions as

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{1.9}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad i \in \{0,1,2,3\} \tag{1.10}$$

where $i \equiv i \pmod 4$ as usual, and $R_{i,i+2} = -R_{i+2,i}$ according to the equation (1.8). Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.3, we draw R_0 on a general quadrilateral element, which evaluates 1 on the edge e_0, and evaluates 0 on the opposite edge e_2.

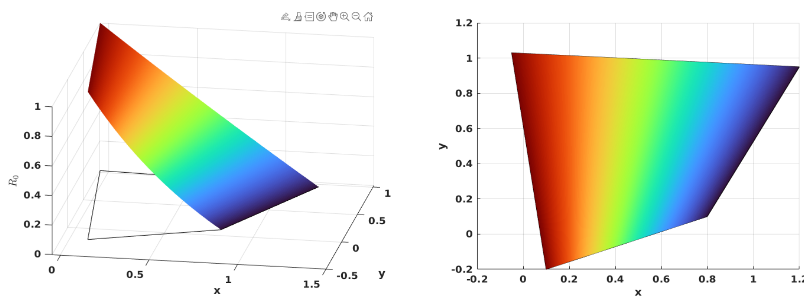


Figure 1.3: Rational function R_0.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{e,i}^{(2)}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.11}$$

where $\mathbf{x}_{e,i}$ is the coordinates of the middle point of the edge e_i. Here, $i \equiv i \pmod 4$. We can restrict the edge basis function to edges as

$$\varphi_{e,i}^{(2)}|_{e,j} = \begin{cases} \dfrac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})}, & i = j \\ 0, & i \neq j \end{cases} \tag{1.12}$$

In Figure 1.4, we present $\varphi_{e,0}^{(2)}$ on a general quadrilateral element, where $\varphi_{e_0}^{(2)}$ evalues 0 on the edges $\{e_i, \quad i \neq 0\}$, and evaluates a quadratic bubble function on the edge e_0.

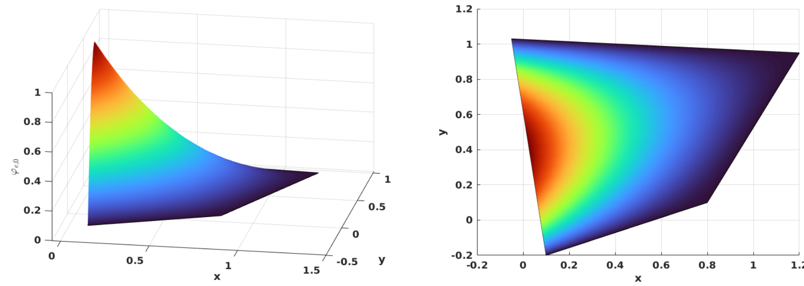


Figure 1.4: Edge nodal basis function $\varphi^2_{e,0}$ in $\mathcal{DS}_2$.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^2_{v,i}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x}_{v,i})}, \tag{1.13}$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i. The i index is understood in the sence of module of 4 as well. In Figure 1.5, we show $\varphi^2_{v,0}$ on a general quadrilateral.

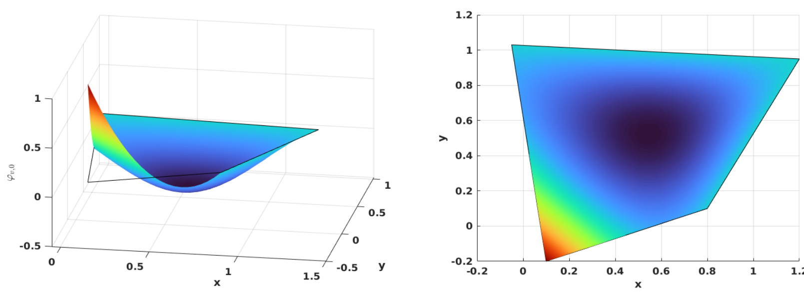


Figure 1.5: Vertex nodal basis function $\varphi^2_{v,0}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.2.2 Direct Serendipity Space $\mathcal{DS}_1$

The supplemental space for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order serendipity as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \mathrm{span}\{R\}, \tag{1.14}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of rational funcion R is not unique. We present one way to construct $R(\mathbf{x})$ with edge nodal basis function (1.11) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x}), \tag{1.15}$$

where auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{1.16}$$

In Figure 1.6, we plot the rational function on a general quadrilateral.

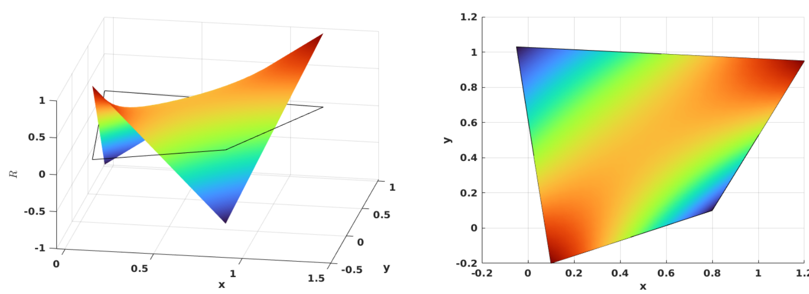


Figure 1.6: Rational function in $\mathcal{DS}_1$.

We define vertex nodal basis function accordingly as

$$\varphi^1_{v,i}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{1.17}$$

where $m \equiv i + 1 \pmod 2$, and index i is understood in the sence of modulo of 4. In Figure 1.7, we show the vertex nodal basis function $\varphi^1_{v,0}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertex.

We first define an interpolation operator of function $v \in L^2(\Omega)$ onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$. Here, K_i has different meanings with different types of basis functions. For each basis functions associated with edge node, we set $K_i = e_i$. For each basis functions associated with vertex node, we set K_i being a set of edges sharing the

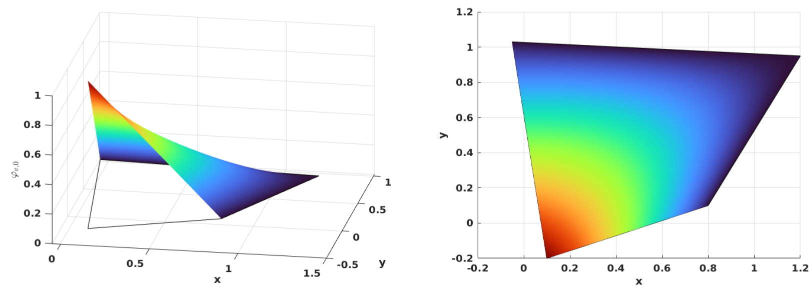


Figure 1.7: Vertex nodal basis function in $\mathcal{DS}_1$.

vertex v_i. Following the work by Arbogast et al. (2022), we define a interpolation operator $\mathcal{I}_h^r : W_p^l(E) \to \mathcal{DS}_r$ as

$$\mathcal{I}_h^r v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y}) d\mathbf{y} \in \mathcal{DS}_r, \tag{1.18}$$

where $1 \leq p \leq \infty$ and $l > 1/p$ (but $l \geq 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x}) \varphi_j(\mathbf{x}) d\mathbf{x} = \delta_{i,j}, \quad i, j \in \{0, 1, 2, \dots, N_r\}, \tag{1.19}$$

where N_r is the number of nodal basis defined on K_i. We present the following two lemmas regarding the constructed interpolation operator for future reference.

Lemma 1.1. *(Boundedness of the interpolation operator.)*

Let $\mathcal{T}_h$ be uniformly shape regular (1.2) *with shape regularity parameter σ_*. For each element $E \in \mathcal{T}_h$, There exists a bounded interpolation operator* (1.18)*. For every $E \in \mathcal{T}_h$, suppose that the basis functions of $\mathcal{DS}_r(E)$ are constructed using λ_{24} and λ_{13} such that the zero set $\mathcal{Z}_{i,i+2}$ intersects e_{i+3} and e_{i+1}. Moreover, assume that $R_{i,i+2}$ is uniformly differentiable functions of the vertices of E up to order m. Then for $r \geq 1$, $E \in \mathcal{T}_h$, $1 \leq q \leq \infty$, and any nonnegative integer m,*

$$\|\mathcal{I}_h^r v\|_{W_q^m} \leq C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \tag{1.20}$$

where $E^* = \cup_{F \in \mathcal{T}_h, F \cap E \neq \emptyset}$ and $|\cdot|_{W_p^k}$ is the seminorm of k-th order derivatives.

Proof. See Arbogast et al. (2022). □

Lemma 1.2. *(Approximability of the interpolation operator.)*

Under the assumptions of 1.1, there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \le p \le \infty$ and $l > 1/p$ or ($l \ge 1$ if $p = 1$),

$$\|v - \mathcal{I}_h^r v\|_{W_p^m} \le C h_E^{l-m} |v|_{W_p^l(E^*)}, \quad 0 \le m \le \min(l, r+1). \tag{1.21}$$

Morevoer, there exists a constant $C = C(r, \sigma_*) > 0$, independent of $h = \max_{E \in \mathcal{T}}$, such that for all functions $v \in W_p^l(\Omega)$,

$$\Big(\sum_{E \in \mathcal{T}_h} \|v - \mathcal{I}_h^r v\|_{W_p^m}^p\Big)^{1/p} \le C h^{l-m} |v|_{W_p^l(\Omega)}, 0 \le m \le \min(l, r+1). \tag{1.22}$$

Proof. See Arbogast et al. (2022). □

1.3 Mixed Finite Element Discretization

Recall Ciarlet's definition Ciarlet (2002) of a finite element.

Definition 1.2. A triple $(E, X(E), \varphi_j)$ defines a finite element.

- Element $E \in \mathcal{T}_h$;
- Space of shape function $X(E)$, and a set of shape function $\varphi_i \in X(E)$. The dimension of the shape function space $\dim(X(E)) = n$;
- A set of continuous and linear functionals ϕ_j

  $$\phi_j : X(E) \supset \mathcal{X}(E) \to \mathbb{R}, \tag{1.23}$$

  such that unisolvence condition is achieved as

  $$\langle \phi_j, \varphi_i \rangle = \delta_{ij}, \quad i, j \in \{1, 2, 3, \ldots, n\} \tag{1.24}$$

In the following sections, we state our shape function space constructed using first and second order direct serendipity space and corresponding degrees of freedom. We also provide explicit construction of shape functions.

We introduce a H^1 and a $H(\mathrm{div})$ conforming mixed element space constructed with the pre-defined direct serendipity space for Darcy and Stokes problems.

1.3.1 A Darcy Model Problem

Consider a model problem for Darcy equation as

$$\begin{aligned} \mathbf{u} + \nabla p = \mathbf{f}, &\ \text{ in } \Omega \\ \nabla \cdot \mathbf{u} = 0, &\ \text{ in } \Omega \\ \mathbf{u} = \mathbf{0}, &\ \text{ on } \partial\Omega \end{aligned} \tag{1.25}$$

We seek solution in the following energy space as

$$\begin{aligned} \mathbb{V}_D &= \{\mathbf{v} \in (H(\mathrm{div};\Omega))^2 : \mathbf{v} \cdot \boldsymbol{\nu} = 0 \ \text{ on } \partial\Omega\}, \\ \mathbb{U}_D &= \{\mathbf{u} \in (H(\mathrm{div};\Omega))^2 : \mathbf{u} \cdot \boldsymbol{\nu} = \mathbf{0} \ \text{ on } \partial\Omega\} = \mathbb{V}_D, \\ \mathbb{W}_D &= \{p \in L^2(\Omega) : \int_\Omega p d\mathbf{x} = 0\} = L^2(\Omega)/\mathbb{R}. \end{aligned} \tag{1.26}$$

We write weak form with these energy space, and consider the standard variational form : find $\mathbf{u} \in \mathbb{U}_D$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\mathbf{u}, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}), &\ \ \forall \mathbf{v} \in \mathbb{V}_D, \\ (\nabla \cdot \mathbf{u}, \omega) = 0, \qquad &\ \ \forall \omega \in \mathbb{W}_D. \end{aligned} \tag{1.27}$$

We introduce two finite-dimensional subspace $\mathbb{V}_{D,h} \subset \mathbb{V}$ and $\mathbb{W}_{D,h} \subset \mathbb{W}$, and consider the discretized problem: find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ such that

$$\begin{aligned} (\mathbf{u}_h, \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) = (\mathbf{f}, \mathbf{v}_h), &\ \ \forall \mathbf{v}_h \in \mathbb{V}_{D,h}, \\ (\nabla \cdot \mathbf{u}_h, \omega_h) = 0, \qquad &\ \ \forall \omega_h \in \mathbb{W}_{D,h}. \end{aligned} \tag{1.28}$$

Now we construct a pair of discretized space $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ for this Darcy model problem in the next section.

1.3.2 A reduced $H(\mathbf{div})$ conforming space $\mathbb{V}_1^0$

We have some choice for $H(\text{div})$ confirming space. For example, we can choose the famous Ravier-Thomas element by Raviart and Thomas (1977). The serendipity space is the precursor of the Brezzi-Douglas-Marini element by Brezzi et al. (1985). We recover Arbogast-Correa element by Arbogast and Correa (2016) if we use mapped direct serendipity space. We generated a reduced $H(\text{div})$ approximation, that $\nabla \cdot \mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$.

We write the rotated 2D exact sequence for energy space as

$$H^1 \xrightarrow{\text{curl}} H(\text{div}) \xrightarrow{\text{div}} L^2 \tag{1.29}$$

$$\cup \qquad\qquad \cup \qquad\qquad \cup \tag{1.30}$$

$$\mathcal{DS}_2 \xrightarrow{\text{curl}} \mathbf{V}_r^{r-1} \xrightarrow{\text{div}} \mathbb{P}_{r-1}. \tag{1.31}$$

We define the first order reduced $H(\text{div})$ conforming space with $\mathcal{DS}_2$ as

$$\mathbb{V}_1^0 = \mathbb{P}_1^2 \oplus \text{curl}\mathbb{S}_2^{\mathcal{DS}}. \tag{1.32}$$

The reduced space $\mathbb{V}_1^0$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements. The dimension of the derived space is

$$\dim(\mathbb{V}_1^0) = 2 \times 3 + 2 = 8. \tag{1.33}$$

We need two independent basis functions on each edge of a quadrilateral E, one of which has a vanishing average flux on the edge and the other one has a non-vanishing average flux on the edge.

We state the degree of freedom of element as normal flux on each edge as

$$\int_{e_i} \mathbf{u} \cdot \boldsymbol{\nu} \varphi_i d\mathbf{x}, \quad \varphi_i \in \mathbb{V}_1^0. \tag{1.34}$$

We need two independent shape functions on each edge. We present a construction of two independent basis functions on each edge: $\boldsymbol{\varphi}_{l,i}$ and $\boldsymbol{\varphi}_{c,i}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \tag{1.35}$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{1.36}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i ds = 0$.

To begin with, we write curl of the auxiliary rational function R_i (1.8) as

$$\mathrm{curl} R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{1.37}$$

where the index i is understood with modulo of 4. We first present a construction of basis function $\boldsymbol{\varphi}_{l,i}$ as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \mathrm{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\mathrm{curl} R_i. \tag{1.38}$$

We then normalize it such that l_i varying from -1 to 1 as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.39}$$

where the index i is understood in the sense of modulo 4, and edge length $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 1.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing a edge, and show normal flux joining continuously as a linear function on the shared edge.

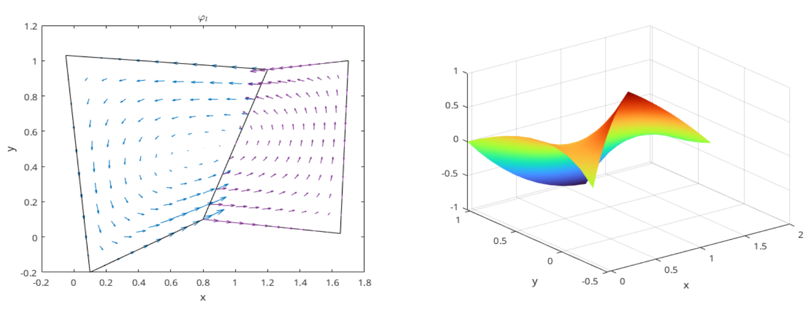


Figure 1.8: Linear basis function $\boldsymbol{\varphi}_l$.

The construction of a the constant basis function is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal function $\phi^*_{v,i} \in \mathcal{DS}_2(E)$ such that $\phi^*_{v,i}(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi^*_{v,i}(\mathbf{x}) = \varphi^2_{v,i}(\mathbf{x}) + \big[\frac{1}{2}\varphi^2_{e,i} + \frac{1}{2}\varphi^2_{e,i+1}\big], \tag{1.40}$$

where the index i is understood in modulo of 4. Then define $\boldsymbol{\varphi}^*_{v,i}(\mathbf{x}) = \mathrm{curl}\phi^*_{v,i}$ so that

$$\boldsymbol{\varphi}^*_{v,i} \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i+1, \\ 0, & j = i+2, i+3. \end{cases} \tag{1.41}$$

We then define the "constant" basis function defined on the edge as

$$\boldsymbol{\varphi}_{c,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|\boldsymbol{\varphi}^*_{v,i} + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \tag{1.42}$$

where the index i is again understood in the sense of modulo of 4. The defined basis function $\boldsymbol{\varphi}_{c,i}$ has flux of constant value 1 on e_i and 0 on all the other edges. In Figure 1.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing a edge, and show normal flux joining continuously as a constant 1 on the shared edge.

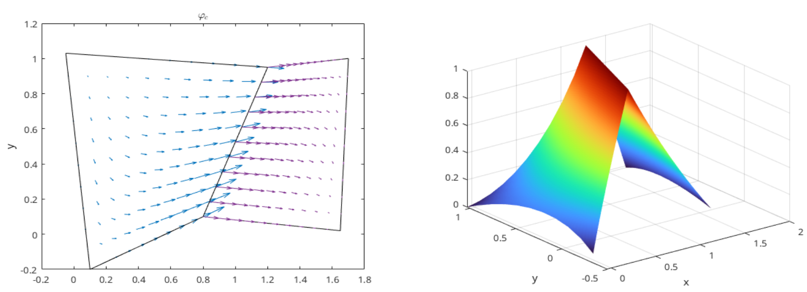


Figure 1.9: Constant basis function $\boldsymbol{\varphi}_c$.

1.3.3 Stability and Approximation Properties for $\mathbb{V}^0_1$

The accuracy and stability of such element has been well studied and tested in the work by Arbogast et al. (2022). We state the result directly for completeness. Consider a projection operatior π_D preserves element average divergence and average edge normal fluxes.

$$\pi_D : H(\mathrm{div};\Omega) \cap (L^{2+\epsilon}(\Omega))^2 \to \mathbb{V}^0_1, \tag{1.43}$$

where $\epsilon > 0$. The global projection operator π is understood as combination of local operator π_E according to degree of freedom. The operator π also satisfies the commuting diagram property as

$$\mathcal{P}_{W_s} \nabla \cdot \mathbf{v} = \nabla \cdot \pi \mathbf{v}, \tag{1.44}$$

where $\mathcal{P}_{W_s}$ is the L^2-orthogonal projection operator onto $W_s = \nabla \cdot \mathbb{V}_1^0$, which is a constant value in our case.

Lemma 1.3. *Stability and approximation properties.*

Under the assumption of 1.1, there exists a constant $C > 0$, independent of $\mathcal{T}_h$ and $h > 0$, such that

$$\|p - p_h\|_{m,\Omega} \leq C h^{l+1-m} |p|_{l+1,\Omega}, \quad l = 0, 1, 2, \ldots, r, \tag{1.45}$$

where $p_h \in \mathcal{DS}_2$. We have

$$\begin{aligned} \|\mathbf{u} - \pi_D \mathbf{u}\|_{0,\Omega} &\leq C \|\mathbf{u}\|_{k,\Omega} h^k, \qquad k = 1, \ldots, 2 \\ \|\nabla \cdot (\mathbf{u} - \pi_D \mathbf{u})\|_{0,\Omega} &\leq C \|\nabla \mathbf{u}\|_{k,\Omega} h^k, \quad k = 0, \ldots, 1 \\ \|p - \mathcal{P}_{W_s} p\|_{0,\Omega} &\leq C \|p\|_{k,\Omega} h^k, \qquad k = 0, \ldots, 1 \end{aligned} \tag{1.46}$$

Moreover, the discrete inf-sup condition holds

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^0} \frac{(\omega_h, \nabla \cdot \mathbf{v}_h)}{\|\mathbf{v}_h\|_{H(div)}} \geq \gamma \|\omega_h\|_{0,\Omega}, \quad \forall \omega_h \in W_s, \tag{1.47}$$

holds for some $\gamma > 0$ independent of $\mathcal{T}_h$ and $h > 0$.

Proof. See Arbogast et al. (2022). □

1.3.4 A Stokes Model Problem

Consider a model problem for Stokes equation as Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{0} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.48}$$

We define a simple choice of energy space

$$\begin{aligned}
\mathbb{V}_S &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \;\text{ on }\; \partial\Omega\} = (H_0^1(\Omega))^2,\\
\mathbb{U}_S &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{0} \;\text{ on }\; \partial\Omega\} = \mathbb{V}_S,\\
\mathbb{W}_S &= \{p \in L^2(\Omega) : \int_\Omega p d\mathbf{x} = 0\} = L^2(\Omega)/\mathbb{R},
\end{aligned} \tag{1.49}$$

We write weak form with these energy space, and consider the standard variational form : find $\mathbf{u} \in \mathbb{U}_S$ and $p \in \mathbb{W}_S$ such that

$$\begin{aligned}
(\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}), \;\; &\forall \mathbf{v} \in \mathbb{V}_S,\\
(\nabla \cdot \mathbf{u}, \omega) = 0, \qquad\quad &\forall \omega \in \mathbb{W}_S.
\end{aligned} \tag{1.50}$$

We introduce two finite-dimensional subspace $\mathbb{V}_{S,h} \subset \mathbb{V}$ and $\mathbb{W}_{S,h} \subset \mathbb{W}$, and consider the discretized problem: find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ such that

$$\begin{aligned}
(\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) = (\mathbf{f}, \mathbf{v}_h), \;\; &\forall \mathbf{v}_h \in \mathbb{V}_{S,h},\\
(\nabla \cdot \mathbf{u}_h, \omega_h) = 0, \qquad\quad &\forall \omega_h \in \mathbb{W}_{S,h}.
\end{aligned} \tag{1.51}$$

Now we construct a pair of discretized space $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ for this Stokes model problem in the next section.

1.3.5 Modified Bernardi-Raugel Element

Stable mixed finite element pair for Stokes problem has been widely researched. We are referring to the element discussed by Christine Bernardi (1985) and by Fortin (1981). A similar type of element was discussed by Arbogast and Wheeler (2005).

Inspired by the work in direct serendipity space, we consider the following modified space for each element $E \in \mathcal{T}_h$ as

$$V_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \text{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \tag{1.52}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined on each edge e_i. One natural pick is the previously defined function $\varphi_{e,i}^{(2)} \in \mathcal{DS}_2$ as

$$\mathbf{b}_i = \varphi_{e,i}^{(2)}(\mathbf{x})\mathbf{n}_i, \tag{1.53}$$

which is quadratic when restricted to edge e_i according to equation (1.12). The quadratic bubble is guaranteed to be conforming since it evaluates zeros on two vertex and 1 at the middle point of the edge due to normalization. The integral of this quadratic bubble can be accurately computed by a three points gaussian quadrature rule.

Definition 1.3. For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following degrees of freedom:

(i) for corner point v_i and Cartesian direction unitnormal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF^{(\mathrm{i})}_{(v_i,\mathbf{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF^{(\mathrm{i})}_{(v_i,\mathbf{j})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) for edge e_i,

$$DOF^{(ii)}_{e_i} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle,$$

We give an explicit construction of basis functions using driect serendipity space. We pick nodal basis on vertex v_i in two Cartesian direction as

$$N_{v_i,x} = \begin{bmatrix} \psi_{v,i}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \psi_{v,i}(\mathbf{x}) \end{bmatrix} \tag{1.54}$$

We pick edge basis on edge e_i as

$$N_e = \varphi^{(2)}_{e,i} \boldsymbol{\nu}_i \tag{1.55}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i, therefore bubble functions cannot be in the same space as nodal basis function. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function.

We define the corresponding discrete pressure sapce as

$$\mathbb{W}_0 = \{\omega \in L^2\Omega; \omega|_E \in P_0\} \cap L^2(\Omega)/\mathbb{R}, \tag{1.56}$$

which holds piece wise constant values for each element E. We provide stability and approximation analysis for this space pair $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ in the next section.

1.3.6 Stability and Approximation Properties for $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$

Before analyze the stability and approximation properties for the modified, we state some useful lemma.

Lemma 1.4. *Let K be a nondengenate convex quadrilateral. Assuming we have shape regularity 1.1. For each integer $m \geq 0$ and real number $p \in [1, \infty]$ there exist positive constants C_1, and C_2 independent of the geometry of E such that the following upper bound hold for all $v \in W_p^m(E)$:*

$$\begin{cases} |v|_{W_p^0(K)} \leq C_1 h_E^{2/p} |\tilde{v}|_{W_p^0(\tilde{K})}, \\ |v|_{W_p^1(K)} \leq C_2(\sigma^*, p) |\tilde{v}|_{W_p^1(\tilde{K})}, \end{cases} \tag{1.57}$$

where $\tilde{K}$ represents quadrilateral before scaling illustrated in Figure 1.1, and $\tilde{v}$ is function defined on $\tilde{K}$ before scaling.

Proof. See Girault and Raviart (2011). □

With the interpolation operator $\mathcal{I}_{\mathcal{DS}_1}$ defined in 1.18, we define the projection operator $\pi_h \in (H_0^1(\Omega))^2 \to \mathbb{V}_1^{BR} \cap (H_0^1(\Omega))^2$ by:

(i) For each corner point $v \in \partial E$

$$\pi_h \mathbf{v}(\mathbf{x}_v) = \mathcal{I}_{\mathcal{DS}_1} \mathbf{v}(\mathbf{x}_v). \tag{1.58}$$

(ii) On each edge $e \subset \partial E$,

$$\int_e (\pi_h \mathbf{v} - \mathbf{v}) \cdot \boldsymbol{\nu} ds = 0. \tag{1.59}$$

Lemma 1.5. *For all $\mathbf{v} \in (H^k(\Omega))^2$, $\quad k = 1, 2$. The operator π_h defined by* (1.58) *and* (1.59) *satisfies:*

$$(p_h, \nabla \cdot (\mathbf{v} - \pi_h \mathbf{v})) = 0, \quad \forall p_h \in \mathbb{W}_0 \tag{1.60}$$

Assuming shape regularity defined in 1.1, π_h should have the error bound as

$$\|\mathbf{v} - \pi_h \mathbf{v}\|_{1,\Omega} \leq C h^m \|\mathbf{v}\|_{m+1,\Omega}, \quad m = 0, 1. \tag{1.61}$$

Proof. From the definition of degree of freedom (1.58) and (1.59), we decompose the projection operator π_h as

$$\pi_h \mathbf{v} = \mathcal{I}_{\mathcal{DS}_1}\mathbf{v} + \sum_{i=0}^{3} \alpha_i \mathbf{b}_i, \tag{1.62}$$

where the coefficient

$$\alpha_i = \Big(\int_{e_i} (\mathbf{v} - \mathcal{I}_{\mathcal{DS}_1}\mathbf{v}) \cdot \boldsymbol{\nu}_i d\sigma \Big) / \int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i d\sigma. \tag{1.63}$$

We use the approximation property of interpolation operator in direct derendipity space in Lemma 1.2,

$$\|\mathbf{v} - \mathcal{I}_{\mathcal{DS}_1}\mathbf{v}\|_{m,\Omega} \le h_E^{2-m} |\mathbf{v}|_{2,E^*}, \quad m = 0, 1, \tag{1.64}$$

where $E^* = \Cup_{F \in \mathcal{T}_h, F \cap E \neq}$ defined in Lemma 1.1.

Lemma 1.4 indicates :

$$|\mathbf{b}_i|_{m,E} \le C(\sigma^*) h_E^{1-m} |\tilde{\mathbf{b}}_i|_{1,\tilde{E}} \le C(\sigma^*) h_E^{1-m}, \quad m = 0, 1. \tag{1.65}$$

We continue to estimate $|\alpha_i|$ as

$$\begin{aligned} \int_{e,i} \varphi_{e,i}^{(2)} ds = |e_i| \int_{\tilde{e}_i} \tilde{\varphi}_{e,i}^{(2)} d\tilde{s}, \\ \int_{e,i} (\mathbf{v} - \mathcal{I}\mathbf{v}) \end{aligned} \tag{1.66}$$

□

1.3.7 Test Modified Bernardi-Raugel Element

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem but with a non-homogenous Dirichlet boundary condition:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p = \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega, \\ \mathbf{u} = \mathbf{g} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.67}$$

We define a simple choice of energy space

$$\begin{aligned}
\mathbb{V} &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \ \text{ on } \partial\Omega\}, \\
\mathbb{U} &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g} \ \text{ on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V}, \\
\mathbb{W} &= \{p \in L^2(\Omega)/\mathbb{R}\},
\end{aligned} \tag{1.68}$$

where $\tilde{\mathbf{g}}$ is a finite energy lift. We assume that $\mathbf{g} \in (H^{1/2}(\Omega))^d$ and understand as a continuous extension $\tilde{\mathbf{g}} \in (H^1(\Omega))^d$ from the boundary into the domain. We write the problem in weak form as:

Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

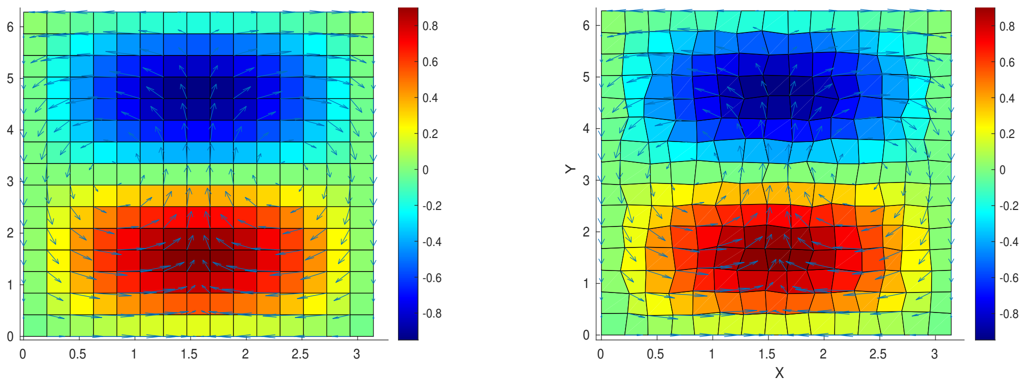
$$\begin{aligned}
(\nabla\mathbf{u}, \nabla\mathbf{v}) - (p, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}), \ \ \forall \mathbf{v} \in \mathbb{V}, \\
(\nabla \cdot \mathbf{u}, \omega) = 0, \qquad \forall \omega \in \mathbb{W}.
\end{aligned} \tag{1.69}$$

We consider the true solution of test problem (1.69) defined on a square region $\Omega = [0, \pi] \times [0, 2\pi]$. We manufacture a displacement and pressure as

$$\mathbf{u} = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \quad p = \sin(x)\sin(y), \tag{1.70}$$

which has 0 normal flux on the boundary. We deal with the constant kernal by enforcing a condition that $\int_\Omega p d\mathbf{x} = 0$ and solve the linear system directly. The numerical results are computed on rectangular 1.10a and logically rectangular quadrilateral 1.10b meshes. We calculate the L^2 norm of the residual (i.e., the difference) of the pressure, velocity, and derivative of velocity.

In Table 1.1 and Table 1.2 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. Second order convergence rate is achieved by velocity and its corresponding derivatives. A first order convergence rate is achieved by pressure. Both meet our expectations for optimal convergence order.



(a) Rectangular Grids. Shown are pressure (color) and velocity (arrows).

(b) Distorted Grids. Shown are pressure (color) and velocity (arrows).

Grid size	8×8	16×16	32×32	64×64
$L^2(\mathrm{res}_u)$	4.502E−1	1.052E−1	2.513E−2	6.130E−3
$L^2(\mathrm{res}_p)$	1.039E+0	4.463E−1	2.093E−1	1.019E−1
$L^2(\mathrm{res}_{du})$	2.632E+0	5.845E−1	1.374E−1	3.330E−2

Table 1.1: Rectangular Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\mathrm{res}_u)$	4.603E−1	1.074E−2	2.567E−2	6.266E−3
$L^2(\mathrm{res}_p)$	1.070E+0	4.670E−1	2.207E−1	1.078E−1
$L^2(\mathrm{res}_{du})$	2.685E+0	5.981E−1	1.525E−1	4.580E−2

Table 1.2: Distorted Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Works Cited

Todd Arbogast and Maicon R. Correa. Two families of h(div) mixed finite elements on quadrilaterals of minimal dimension. *SIAM Journal on Numerical Analysis*, 54(6):3332–3356, 2016. doi: 10.1137/15M1013705. URL `https://doi.org/10.1137/15M1013705`.

Todd Arbogast and Mary F. Wheeler. A family of rectangular mixed elements with a continuous flux for second order elliptic problems. *SIAM Journal on Numerical Analysis*, 42(5):1914–1931, 2005. doi: 10.1137/S0036142903435247. URL `https://doi.org/10.1137/S0036142903435247`.

Todd Arbogast, Zhen Tao, and Chuning Wang. Direct serendipity and mixed finite elements on convex quadrilaterals. *Numerische Mathematik*, 150:929–974, 2022.

Franco Brezzi, Jim Douglas Jr., and L. D. Marini. Two families of mixed finite element for second order elliptic problems. *Numerische Mathematik*, 47:217–235, 1985.

Genevieve Raugel Christine Bernardi. Analysis of some finite elements for the stokes problem. *Math. of Comput.*, 44:71–79, 1985.

Philippe G. Ciarlet. *The Finite Element Method for Elliptic Problems*. Society for Industrial and Applied Mathematics, 2002. doi: 10.1137/1.9780898719208. URL `https://epubs.siam.org/doi/abs/10.1137/1.9780898719208`.

Michel Fortin. Old and new finite elements for incompressible flows. *International Journal for Numerical Methods in Fluids*, 1:347–364, 1981. URL `https://api.semanticscholar.org/CorpusID:122770101`.

Vivette Girault and Pierre-Arnaud Raviart. *Finite Element Methods for Navier-Stokes Equations: Theory and Algorithms.* Springer Publishing Company, Incorporated, 1st edition, 2011. ISBN 3642648886.

P. A. Raviart and J. M. Thomas. A mixed finite element method for 2-nd order elliptic problems. In Ilio Galligani and Enrico Magenes, editors, *Mathematical Aspects of Finite Element Methods*, pages 292–315, Berlin, Heidelberg, 1977. Springer Berlin Heidelberg. ISBN 978-3-540-37158-8.